



ScienceDirect®

Journal of Alloys and Compounds

Volume 1010, 5 January 2025, 177778

Wetting and interface behavior between borosilicate glass and YAG



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<https://doi.org/10.1016/j.jallcom.2024.177778>

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Highlights

- Four borosilicate glasses with good wettability were designed.
- Three types interface behavior between glasses and YAG were revealed.
- The YAG/YAG joint with superior mechanical and optical property was obtained.

Abstract

In this work, four kinds of borosilicate glasses ($\text{La}_2\text{O}_3/\text{SrO}/\text{MgO}/\text{BaO}-\text{B}_2\text{O}_3-\text{SiO}_2$) were designed to investigate the wetting and interface behavior between glass filler and YAG crystal. The thermal property, coefficient of thermal expansion (CTE), crystallization activation energy, optical property and wetting behavior of four kinds of glasses were first studied. The results showed that SBS ($\text{SrO}-\text{B}_2\text{O}_3-\text{SiO}_2$) glass had the most matching CTE with YAG. In addition, all four kinds of glasses had good transparency and exhibited good wettability on the surface of YAG. Furthermore, microstructural analysis of wetting samples indicated that the interface behavior between borosilicate glasses and YAG could be divided into three types: There was only dissolution and diffusion of elements between glass and YAG; The elements of YAG dissolved into glass filler, changing the chemical composition of glass and promoting glass crystallization; Interface reaction occurred between glass and YAG to form the interface layer. Finally, the SBS glass was chosen to bond YAG. The YAG/SBS/YAG joint with high shear strength (78.77 ± 4.3 MPa) and good in-line transmittance (76.5% at 1064 nm) was obtained.



Keywords

YAG; Wetting behavior; Interface behavior; In-line transmittance

1. Introduction

Solid-state lasers have dominated the development of lasers due to their small size, lightweight, simple maintenance and operation, and high output power [1], [2], [3]. At present, YAG single crystal doped with rare earth ions is the most widely used and the most productive solid laser gain medium [4], [5]. The advantages of YAG include high temperature resistance, high-strength, and high-thermal conductivity [6], [7]. It has a stable cubic phase structure without birefringence, making it easy to achieve transparency [4]. However, the cost of YAG single crystal is high and the complex shapes are difficult to prepare [8].

The joining method can not only fabricate large-size or sophisticated shape samples but also reduce the cost of manufacturing. Until now, the methods used to bond transparent ceramics involve diffusion bonding [9], [10], active metal brazing [11], [12] and glass joining [13], [14]. These methods can realize the reliable bonding of transparent ceramics. However, diffusion bonding generally demands high-temperature and high-pressure during joining process, which is not suitable for the fabrication of complex-shaped samples [15]. The active metal filler can reduce the optical properties of the joints due to optical opacity of metal fillers [16]. For joining transparent ceramics, glass joining is determined to be a suitable method on the grounds that glass fillers have good chemical compatibility with ceramics and the coefficient of thermal expansion (CTE) of glass fillers is adjustable, especially the high transparency of glass itself.

For glass joining of transparent ceramics, it is necessary to obtain a completely glassy brazing seam to ensure high transmittance of the joint. The crystals precipitated in the glass filler during heat treatment will result in loss of joint transparency. Lin et al. [17] used bismuth-borate-based glass as filler to join YAG single crystal and the optimal strength of joint reached 29.6 ± 5.2 MPa at 650 °C. The formation of ZnAl_2O_4 and Y_2O_3 nanowires in the brazing seam could increase the strength of the joint, but they will reduce the transparency of the joint. When a rapid cooling rate is applied during joining process, glass crystallization can be inhibited. Fang and co-workers [18] detected that when using $\text{CaO-Li}_2\text{O-Al}_2\text{O}_3\text{-SiO}_2$ glass to join porous Si_3N_4 and dense Si_3N_4 , an amorphous brazing seam of joint was gained with 50 °C/min cooling rate. In our previous work [19], $\text{ZnO-Al}_2\text{O}_3\text{-SiO}_2$ glass was used to join MgAl_2O_4 with a rapid cooling rate of 50 °C/min and an amorphous brazing seam was also gained. Furthermore, the transparency of the joint decreases when the glass filler reacts with ceramic to form an interfacial layer. For instance, the transmittance of the $\text{MgAl}_2\text{O}_4/\text{La}_2\text{O}_3\text{-SiO}_2\text{-B}_2\text{O}_3$ glass/ MgAl_2O_4 joint was only 10.4% at 1000 nm due to the formation of the Al_2O_3 layer at the interface [20]. However, interfacial reaction layer is difficult to remove by changing the parameters of joining process. It can only be removed by adjusting the chemical composition of glass filler itself. Ashizuka and co-workers [21] utilized CaO-MgO-SiO_2 glass to bond zirconia ceramic and the interfacial layer with diopside crystals was found at the joint interface. When 10.5 wt% Al_2O_3 was added to glass filler, an amorphous brazing seam was obtained. Hence, it is crucial to study the interface behavior between glass fillers and transparent ceramics for optimizing the composition of glass fillers and then obtaining joints with high transparency.

In this work, four kinds of borosilicate glass fillers ($\text{La}_2\text{O}_3/\text{SrO/MgO/BaO-B}_2\text{O}_3\text{-SiO}_2$) were designed. The characteristics of four kinds of glasses were studied, and the wettability of different glass fillers on YAG surface was analyzed. The interface behavior between glass fillers and YAG had been revealed. On this basis, the most suitable glass filler was selected for bonding YAG, the mechanical and optical performances of YAG/YAG joints were investigated.

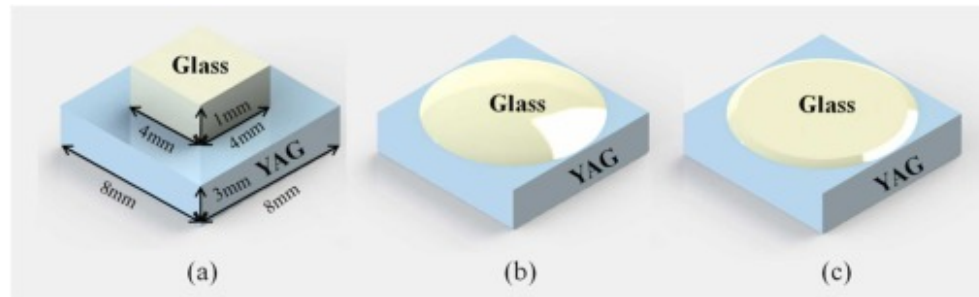
2. Experimental procedures

The YAG crystal was bought from Jiaxing Shengguang Optoelectronic Technology Co., Ltd, Jiaxing, China. YAG with (111) lattice plane was used as the wetting and joining surface. [Table 1](#) displays the composition of four borosilicate glass fillers used in this work. High-purity La_2O_3 , SrCO_3 , MgO , BaCO_3 , H_3BO_3 and SiO_2 powders were purchased from Sinopharm Chemical Reagent Co., Ltd, Shanghai, China. Melting-water quenching technique was used to fabricate the glasses with different compositions, which was labeled as LBS, SBS, MBS and BBS, respectively.

Table 1. Chemical composition of glass fillers (wt%).

Composition	La_2O_3	SrO	MgO	BaO	B_2O_3	SiO_2
LBS	60				10	20
SBS		60			10	20
MBS			60		10	20
BBS				60	10	20

A different scanning calorimeter (DSC, STA449C) was employed to confirm the glass powders' thermal properties. The CTE of different glasses and YAG with [111] crystal orientation was determined by a dilatometer (DIL402PC). The sample size for CTE measurement was $4 \times 4 \times 6 \text{ mm}^3$. The refractive index of glass and ceramics blocks with the size of $8 \times 8 \times 4 \text{ mm}^3$ was measured by an abbe refractometer (Atago, NAR-4T, Japan). The wetting process involved placing the glass block on the surface of YAG and then subjecting them to heat treatment in a tubular furnace at a heating rate of $5 \text{ }^\circ\text{C/min}$ ([Fig. 1a](#)). The degree of spreading of the glass on YAG surface with increasing temperature was took a shot through a charge-coupled device video camera. After the wetting experiment, the glass/YAG samples were obtained by rapid cooling with $50 \text{ }^\circ\text{C/min}$ cooling rate for microstructure analysis ([Fig. 1b](#)). The sample used for in-line transmittance testing required a flat and smooth surface. The upper and lower surfaces of the glass/YAG samples obtained after wetting experiment was smoothed and polished by the precision lapping/polishing machine (UNIPOL-802). The final thickness of the glass layer was 0.2–0.3 mm. The sample used for in-line transmittance testing was shown in [Fig. 1\(c\)](#). The YAG/glass/YAG joint was fabricated by setting one glass sheet between two YAG. The bonding process of YAG/glass/YAG joints was the same as that of wetting experiments of glass/YAG samples.



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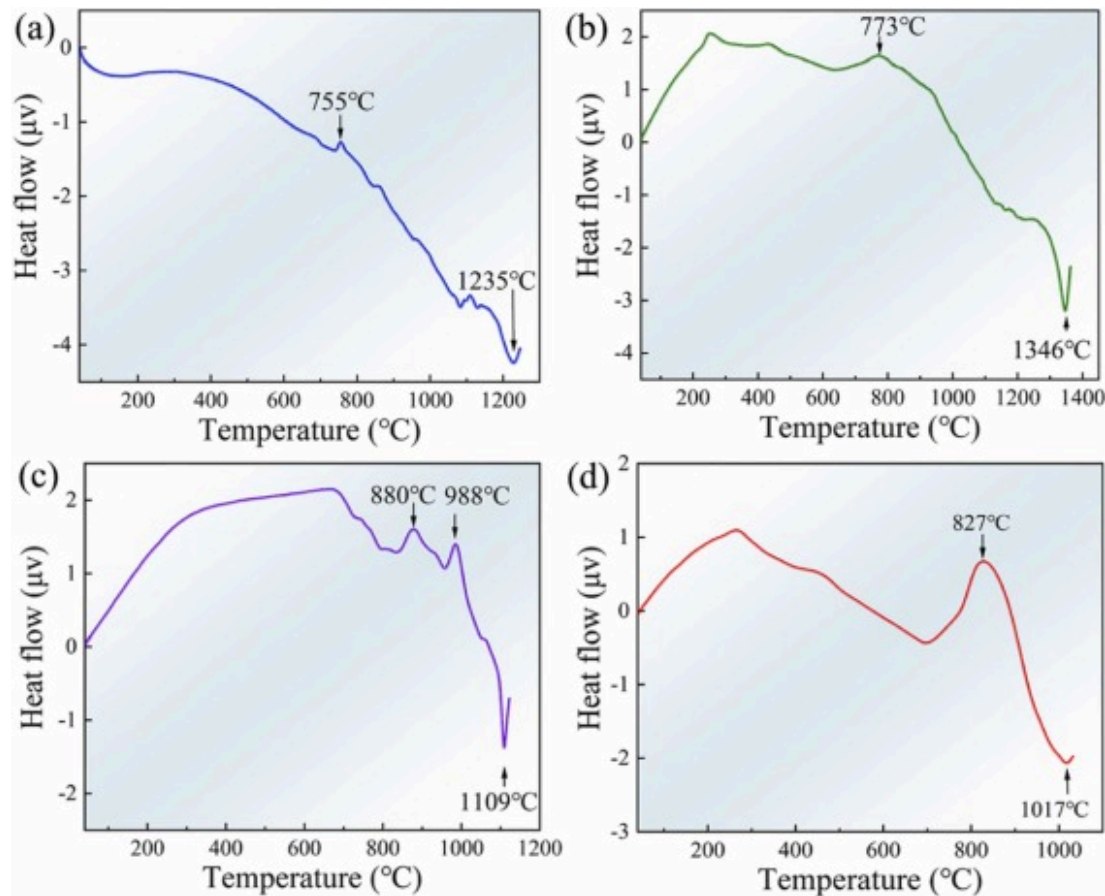
Fig. 1. Sketch of the glass/YAG sample for wetting test (a), microstructure analysis (b), and in-line transmittance test (c).

Scanning electron microscopy (SEM, JSM-IT500A) and energy spectrometry were utilized to analyze the microstructure and elemental distribution of samples. The X-ray diffraction (XRD, Dmax/2500PC) was used to study the phase composition of glass/YAG samples and YAG/glass/YAG joints. The phase composition of glass/YAG samples was investigated by transmission electron microscopy (TEM, Talos F200s) and the sample foil for TEM was prepared using focused ion beam. The universal testing machine (Instron, 34TM-5) was used to measure the shear strength of joints. The ultraviolet-visible (Agilent, Cary 60 UV-Vis) was employed to analyze the in-line transmittance of joints.

3. Results and discussions

3.1. Characterization of glass fillers

The DSC curves of different glasses with 15 °C/min testing rate are displayed in [Fig. 2](#). The exothermic peak in the curve stands for the crystallization peak temperature (T_p) of glass, the endothermic peak stands for the melting temperature (T_m) of crystalline phases. The LBS, SBS and BBS only had one exothermic peak at 755 °C, 773 °C and 827 °C, respectively. However, the MBS glass had two exothermic peaks at 880 °C and 988 °C.



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Fig. 2. DSC curves of glasses at 15 °C/min: (a) LBS glass, (b) SBS glass, (c) MBS glass, (d) BBS glass.

Crystallization activation energy (E) is one of the main parameters in the study of glass kinetics [22]. Crystallization activation energy can be used to judge the stability of glass. E can be obtained through Ozawa [23] formula.

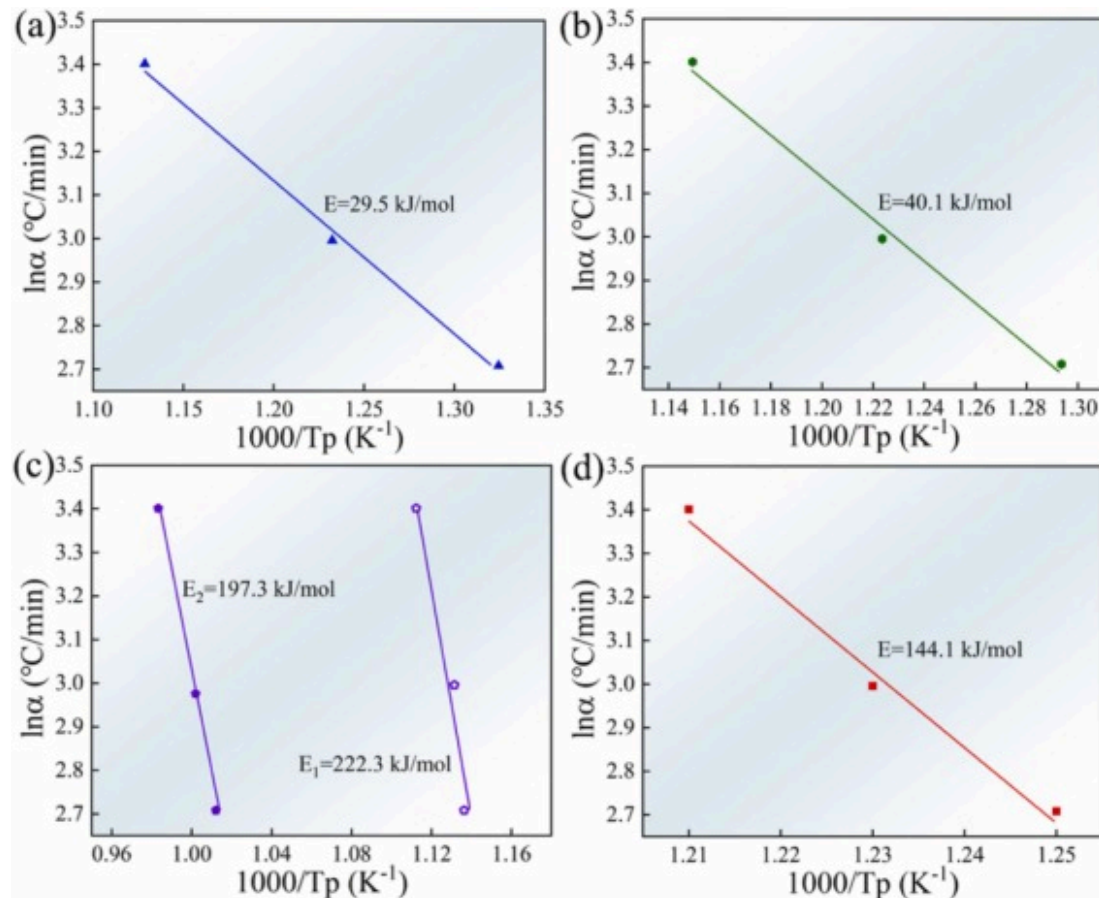
$$\ln \alpha = -E/RT_p + \text{constant} \quad (1)$$

where α is the heating rate of the DSC test, E stands for the crystallization activation energy of glass, T_p represents crystallization

peak temperature of glass and R represents the gas constant. DSC analysis was performed for different glasses with diverse testing rates and the corresponding T_p values obtained for different glasses are shown in Table 2. Based on the changes in T_p reflected from DSC curves of glass to determine E , the results are shown in Fig. 3. The slope of the $\ln\alpha$ and $1000/T_p$ curve reflects the value of E . The E value of LBS glass (29.5 kJ/mol) and SBS glass (40.1 kJ/mol) was lower than that of MBS glass ($E_1=222.3$ kJ/mol, $E_2=197.3$ kJ/mol) and BBS glass (144.1 kJ/mol), indicating that the LBS and SBS glass were easier to crystallize than the MBS and BBS glass.

Table 2. T_p (°C) of different glasses obtained at different heating rates.

Heating rate Samples	15 °C/min		20 °C/min		30 °C/min	
	Tp1	Tp2	Tp1	Tp2	Tp1	Tp2
LBS	755		812		887	
SBS	773		817		867	
MBS	880	988	884	998	899	1027
BBS	827		849		854	



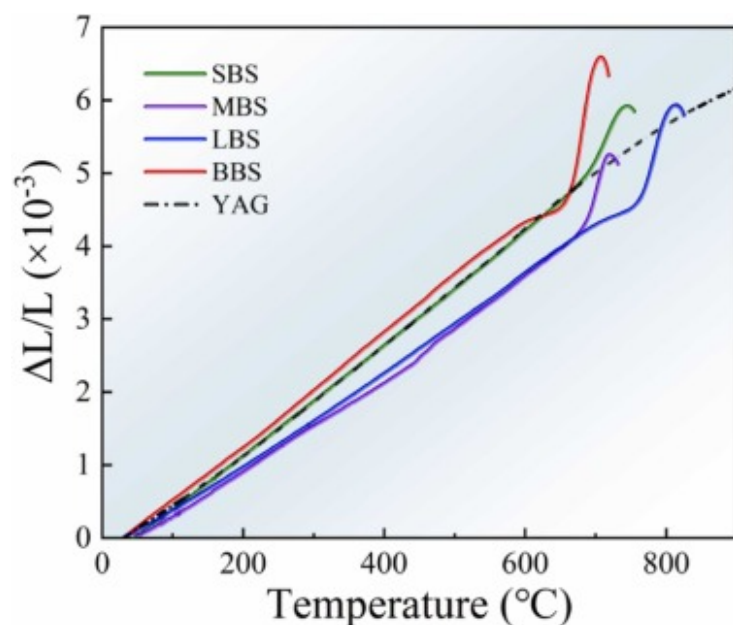
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Fig. 3. Relationship between $\ln\alpha$ and $1000/T_p$ for glasses: (a) LBS glass, (b) SBS glass, (c) MBS glass, (d) BBS glass.

Fig. 4 displays the thermal expansion curves of the different glasses and YAG. It can be seen that all the thermal expansion curves of the glasses show inflection points. The inflection point on the thermal expansion curve of glass is caused by changes in the glass structure. During the heating process, as the glass structure gradually dissociates, the expansion rate of the glass will accelerate within a certain temperature range, resulting in an inflection point on the thermal expansion curve [24]. In general, the slope of the line before the inflection point of the thermal expansion curve represents the CTE of the glass [25]. Therefore, the comparative analysis of the CTE of glasses and YAG only needs to consider the CTE reflected in the temperature range of 30–600

°C, and the CTE of borosilicate glasses and YAG is listed in Table 3. For the amorphous borosilicate glass, the CTE of glass is mainly affected by the field strength of cations. The smaller the field strength of cation, the easier the glass expands, which means the CTE of glass is higher [26]. The field strength of the cation is: Ba^{2+} (0.24) < Sr^{2+} (0.27) < La^{3+} (0.43) < Mg^{2+} (0.45) [27]. Therefore, the lowest and highest CTE values were gained for the MBS glass ($5.75 \times 10^{-6}/^\circ\text{C}$) and BBS glass ($7.59 \times 10^{-6}/^\circ\text{C}$), respectively. In addition, the CTE of SBS glass was $7.38 \times 10^{-6}/^\circ\text{C}$, which was the best match with the CTE of YAG ($7.45 \times 10^{-6}/^\circ\text{C}$). The matched CTE between glass filler and the based substrate is beneficial for obtaining high-strength joints due to the low residual stress [28].



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Fig. 4. Thermal expansion curves of YAG and different glasses.

Table 3. Characteristic temperatures of the glasses, CTE (30–600 °C) and In-line transmittance of glasses and YAG.

Samples	CTE ($\times 10^{-6}/^\circ\text{C}$)	T _g (°C)	T _s (°C)	In-line transmittance@1064 nm
LBS	5.91	761	814	79.1 %

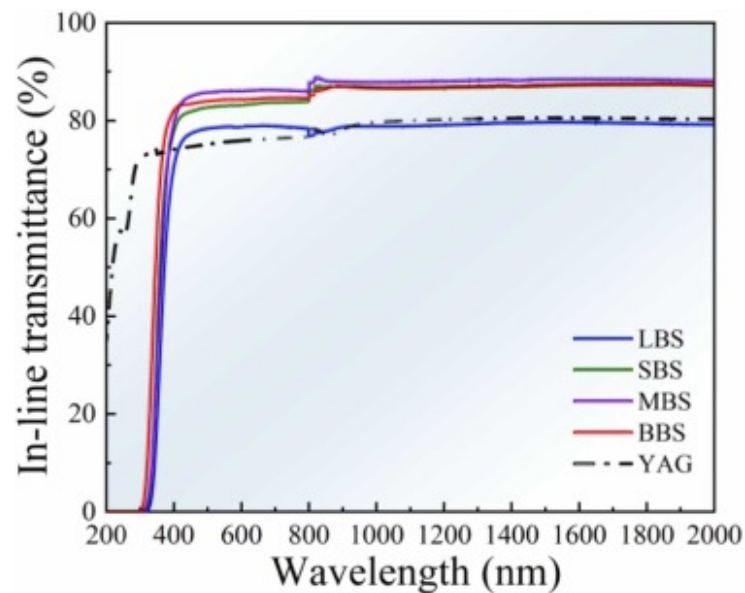
Samples	CTE ($\times 10^{-6}/^{\circ}\text{C}$)	T _g ($^{\circ}\text{C}$)	T _s ($^{\circ}\text{C}$)	In-line transmittance@1064 nm
SBS	7.38	696	744	86.6 %
MBS	5.75	688	719	87.7 %
BBS	7.59	663	708	86.8 %
YAG	7.45	/	/	79.9 %

Fig. 5 illustrates the in-line transmittance of four glasses and YAG. The specific in-line transmittance values at 1064 nm of different glasses and YAG are listed in Table 3. It could be observed that all glasses had excellent in-line transmittance. The transmittance curves fluctuate in the visible to infrared wavelength band (700–800 nm) due to equipment errors. The in-line transmittance of the LBS was similar to YAG, while the in-line transmittance of the other three types of glasses was higher than that of YAG. The in-line transmittance of glass is related to its refractive index. Table 4 lists the refractive indices of different glasses and YAG used in this work. The LBS glass had the highest refractive index due to La_2O_3 having the highest refractive index factor [29]. The refractive index of glass conforms to the additive formula [30]:

$$n = \sum(n_i x_i) \quad (2)$$

where n_i is the refractive index factor of each oxide, x_i is the mole fraction of each oxide. The refractive index factors of different oxides are listed in Table 5. Hence, LBS glass exhibits the highest refractive index because La_2O_3 contributes significantly to its refractive index among all oxides used in this work. When light incidents on the glass surface, the relationship among the reflectivity (R), absorbance (A), and transmittance (T) of the glass conforms to the formula 3 [29]:

$$R + A + T = 100\% \quad (3)$$



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Fig. 5. In-line transmission spectra of YAG and different glasses.

Table 4. The refractive indices of different glasses and YAG (@589.3 nm).

Sample	LBS	SBS	MBS	BBS	YAG
Refractive index	1.7002	1.6952	1.6084	1.5948	1.7989

Table 5. The refractive index factors of different oxides [32].

Oxide	La ₂ O ₃	SrO	MgO	BaO	B ₂ O ₃	SiO ₂
Refractive index factor	2.47	1.78	1.62	1.87	1.46–1.72	1.458–1.175

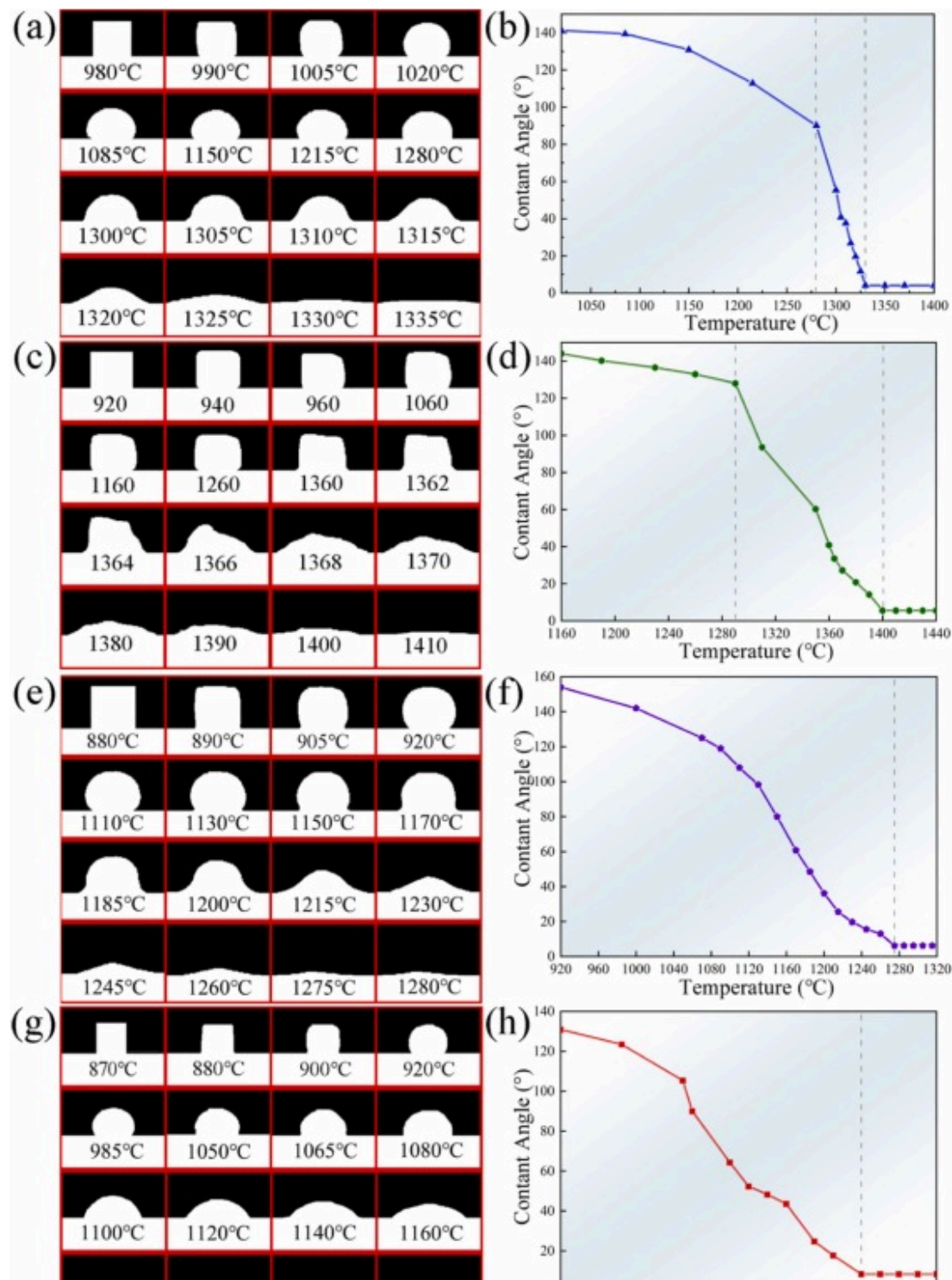
When the incident angle is 90°, the reflectivity (R) can be expressed by formula 4 [29]:

$$R = \left(\frac{n-1}{n+1} \right)^2 \quad (4)$$

According to formula 4, the reflectivity (R) of glass increases with the increase of its refractive index (n). The high refractive index (n) of LBS glass led to its high reflectivity (R) and low transmittance (T). In addition, the LBS glass had the lowest crystallization activation energy (Fig. 2), phase separation may be generated in glass, causing light scattering and resulting in a decrease in the transparency of LBS glass [31].

3.2. Wetting behavior of borosilicate glasses and YAG

It is widely accepted that reliable joints can only be obtained when the glass filler has good wettability with the ceramic [33]. Fig. 6 displays the wetting behavior of the four glasses on YAG. The contact angle of different glass curves as a function of temperature can be divided into two types. The first type was that as temperature increased, the contact angle first slowly decreased, then rapidly decreased, and finally remain stable (Fig. 6(b and d)). The LBS/YAG and SBS/YAG wetting systems belong to the first type. Taking the variation of contact angle of LBS glass on the surface of YAG as an example (Fig. 6a-b), when the temperature was raised to 1020 °C, the glass exhibited spheroidization on account of high surface tension. In the temperature range from 1020 °C to 1280 °C, the contact angle only slightly decreased. The low crystallization activation energy of the LBS and SBS glasses (Fig. 3) promoted crystal precipitation, leading to high viscosity of glass and hindering the spreading of the glass on the substrate [34]. To verify this statement, four borosilicate glasses were heated to 100 °C above their spheroidization temperature and cooled at 50 °C/min, the SEM images shown in Fig. 7. The crystal content in the LBS and SBS glasses was higher than that in the MBS and BBS glasses. When the temperature increased to 1280 °C, higher than T_m of the LBS glass (Fig. 2), the crystals completely melted and the viscosity of the glass significantly decreased with the increasing temperature, leading to a rapidly decreased of contact angle. At last, as the temperature further increased, the contact angle remained unchanged and the equilibrium contact angle of the LBS on YAG surface was approximately 4.1°.

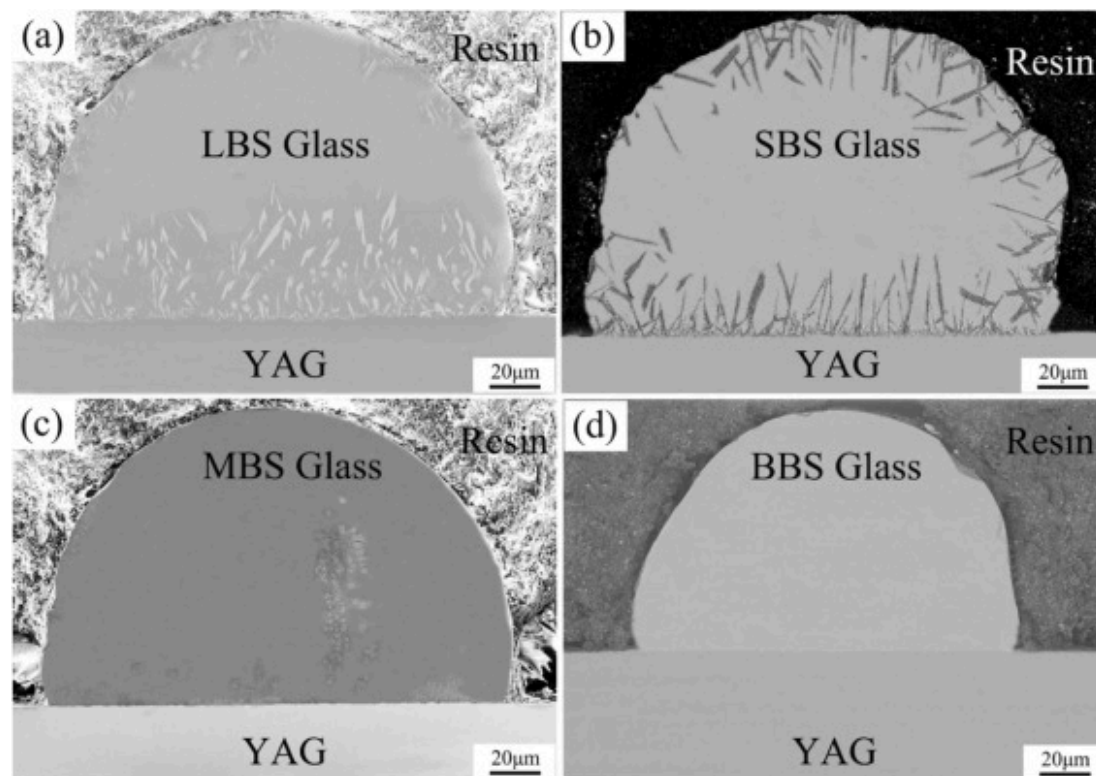




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Fig. 6. Shape changes of different glasses on the surface of YAG: (a) LBS glass, (c) SBS glass, (e) MBS glass, (g) BBS glass. Contact angle of different glasses curves as a function of temperature: (b) LBS glass, (d) SBS glass, (f) MBS glass, (h) BBS glass.



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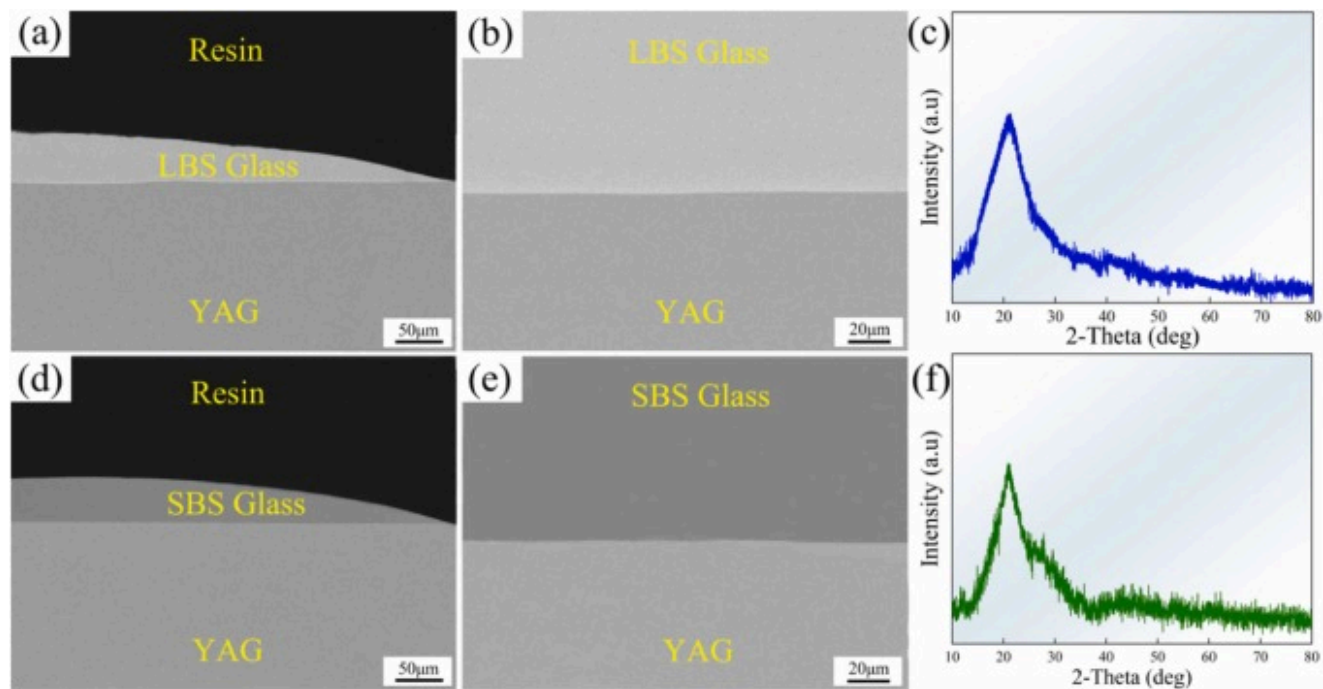
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Fig. 7. SEM images of borosilicate glass/YAG samples: (a) LBS/YAG sample joined at 1120 °C (b) SBS/YAG sample joined at 1160 °C, (c) MBS/YAG sample joined at 1020 °C, (d) BBS/YAG sample joined at 1020 °C.

The second type was that as the temperature increased, the contact angle first decreased almost uniformly and then remained unchanging (Fig. 6(f and h)). The MBS/YAG and BBS/YAG wetting systems belong to the second type. The high crystallization activation energy of MBS and BBS glasses could inhibit glass crystallization. For this type, the changes in contact angle mainly depended on the decrease in the viscosity of glass caused by increasing temperature. Eventually, as the temperature further increased, the contact angle of the MBS/YAG and BBS/YAG wetting systems no longer changed, and remained around 8.3° and 6.2°, respectively. The final contact angles of the four glass fillers on YAG substrate were all less than 10°, indicating that they all had good wettability with YAG.

3.3. Interface behavior of borosilicate glass and YAG

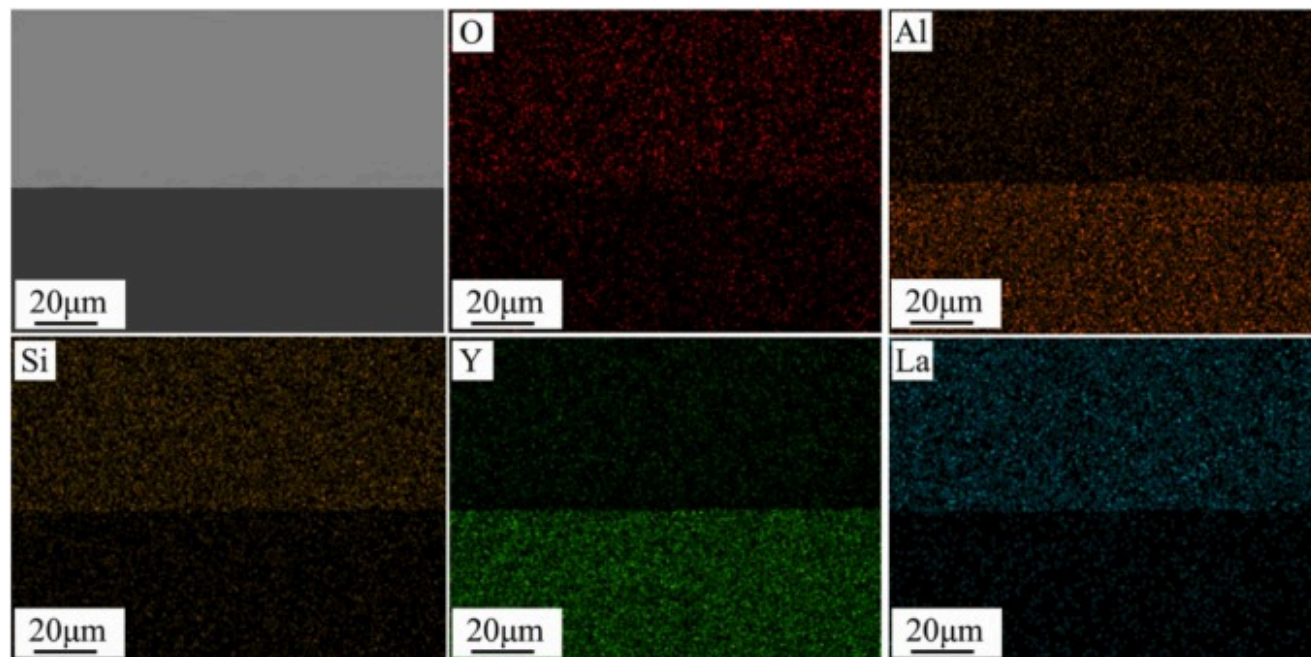
Fig. 8 displays the microstructure of the LBS/YAG samples and SBS/YAG samples. It could be observed from Fig. 8(a-b) and Fig. 8(d-e) that no crystals precipitated at both the LBS/YAG and SBS/YAG interfaces, and the interfaces were relatively smooth. The XRD analysis of the LBS/YAG and SBS/YAG samples (Fig. 8(c and f)) displayed that there was completely amorphous, which illustrated that LBS glass and SBS glass had no interface reaction with YAG. Fig. 9 displays the element distribution maps of the LBS/YAG sample. It can be seen that the Al and Y elements dissolved from YAG into the LBS glass, the elements of La and Si in the LBS glass diffused into YAG, indicating a strong adherence between LBS glass and YAG.



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Fig. 8. SEM images of the LBS/YAG sample (a-b), SBS/YAG sample (d-e). XRD patterns of the LBS/YAG sample (c), SBS/YAG sample (f).

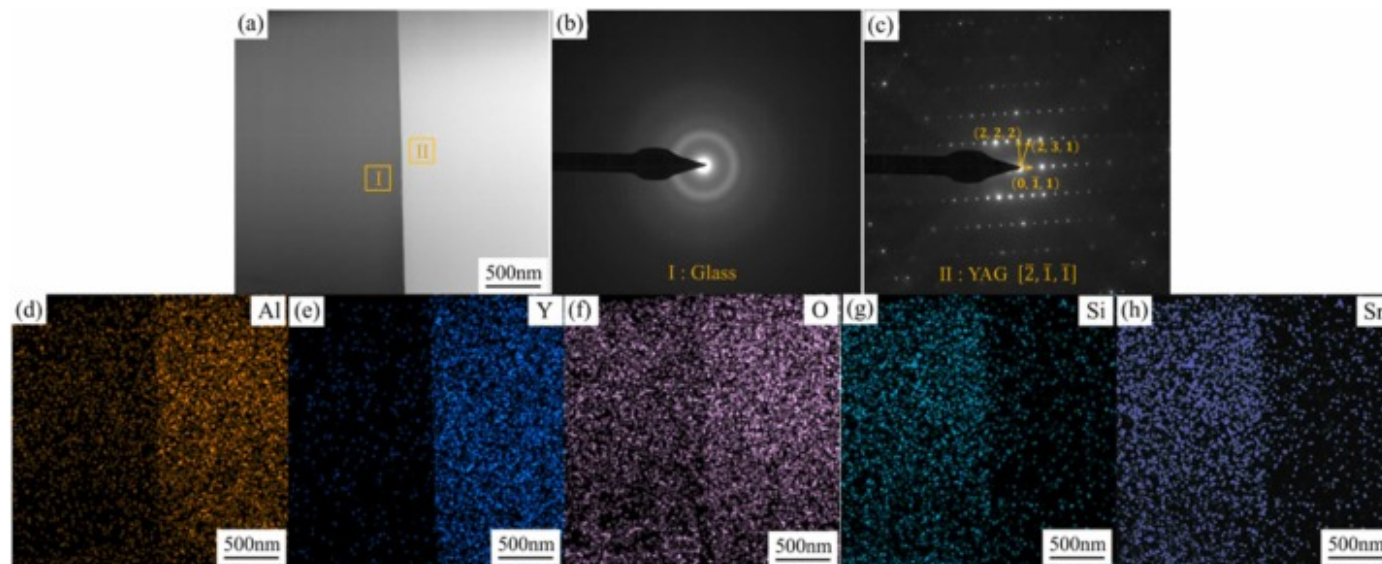


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Fig. 9. Element distribution maps of the LBS/YAG sample.

Due to the CTE of the SBS glass was best matched with that of YAG. The TEM image of the SBS/YAG sample is illustrated in [Fig. 10\(a\)](#). It demonstrated that there was no new phase formed at the interface. [Fig. 10\(b-c\)](#) illustrates the selected area electron diffraction of the SBS/YAG sample, which reconfirms that no crystals formed at SBS/YAG interface. [Fig. 10\(d-h\)](#) displays the element distribution maps of the SBS/YAG sample. It can be seen that the Al and Y elements dissolved from YAG into the SBS glass, the elements of Si and Sr in the SBS glass diffused into YAG, indicating a good interfacial bonding between SBS glass and YAG.

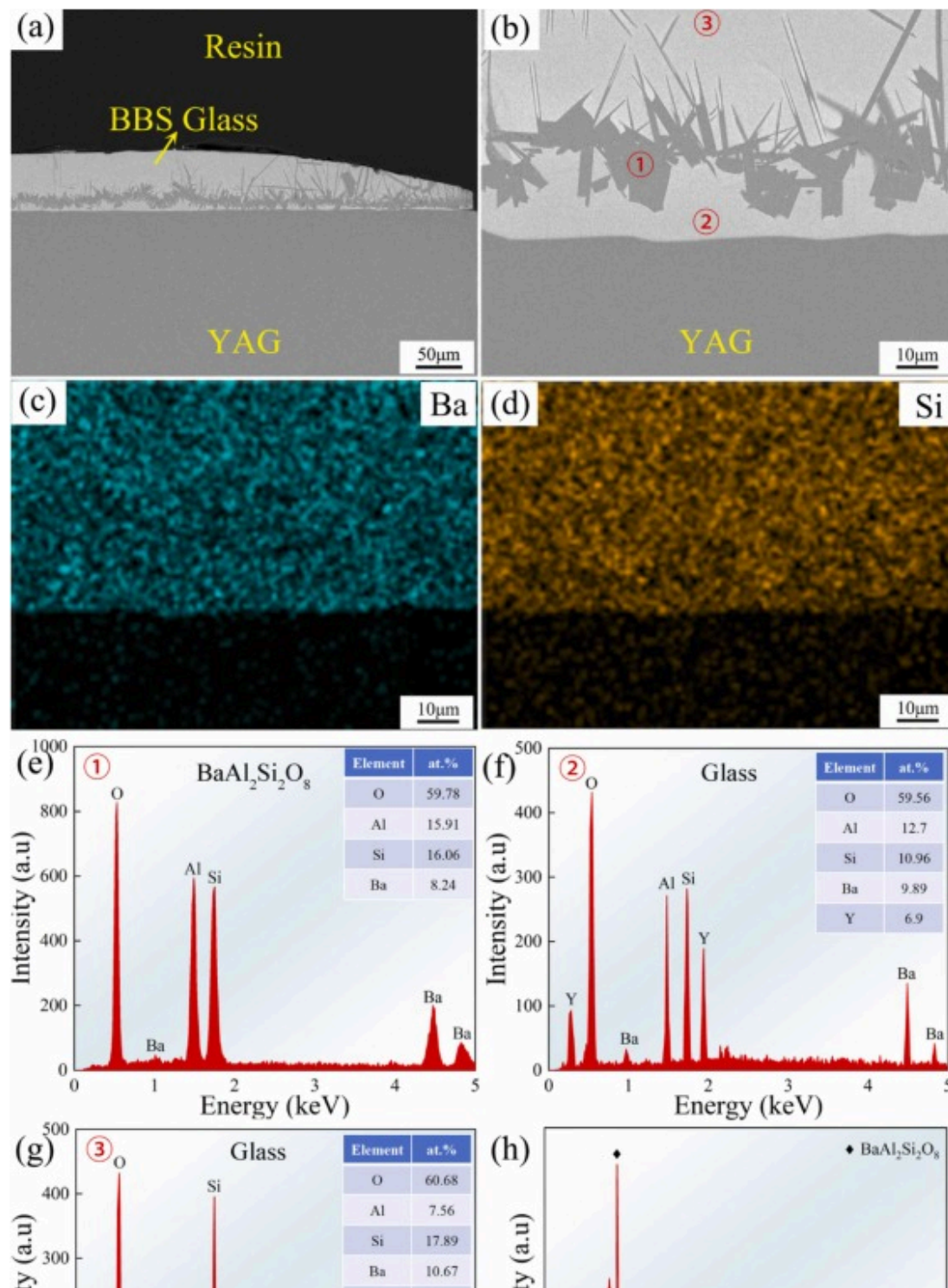


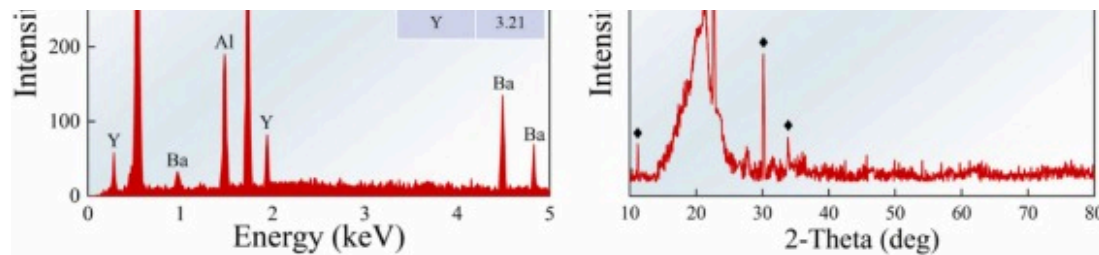
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Fig. 10. TEM image of the SBS/YAG sample (a), (b)–(c) elected area electron diffraction results of areas in (a), element distribution maps (d)–(h) of image (a).

The microstructure of the BBS/YAG sample is displayed in Fig. 11. There are acicular crystals precipitated at the interface of the BBS/YAG sample (Fig. 11(a–b)). Fig. 11(c–d) display the element distribution maps of BBS/YAG sample, it can be seen that a small amount of Ba and Si elements diffused from BBS glass into the YAG substrate. Fig. 11(e–g) show the EDS of point 1–3 in Fig. 11(b), respectively. Combined with the result of the XRD pattern of BBS/YAG sample (Fig. 11(h)), it can be inferred that the precipitated crystal was $\text{BaAl}_2\text{Si}_2\text{O}_8$. It is worth noting that there was no Al element in the original BBS glass composition. The presence of Al element in the precipitated crystals was derived from the dissolution of YAG during the wetting process. The precipitated crystals were at a certain distance from the YAG, which may be due to the high content of the Y element near the YAG side (Fig. 11(d)). Y_2O_3 as a glass network intermediate, has a strong aggregation effect and mainly fills the gaps in the glass network. This characteristic increases the crystallization activation energy of the glass [35] and suppresses the precipitation of crystal near the YAG side.





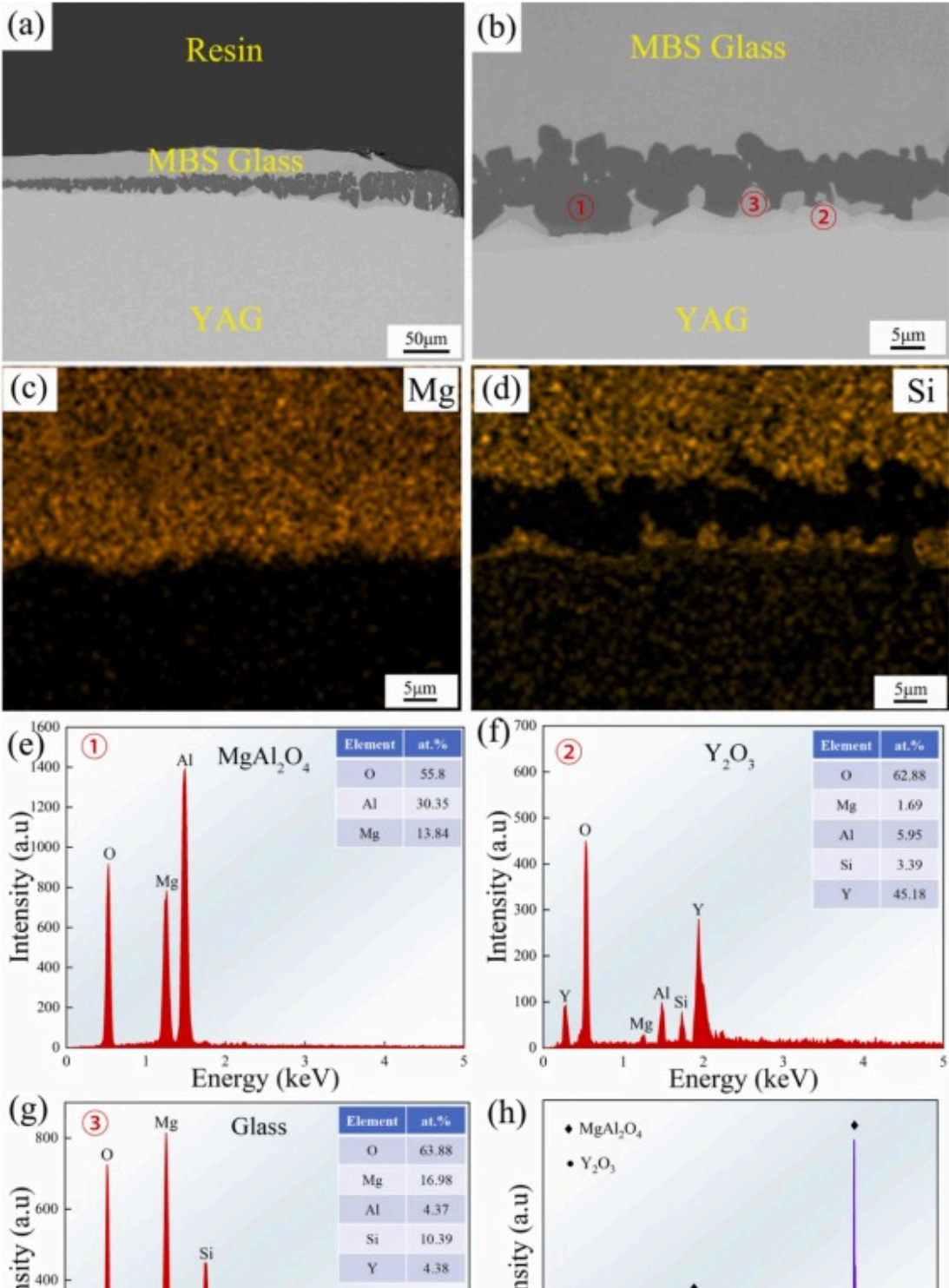
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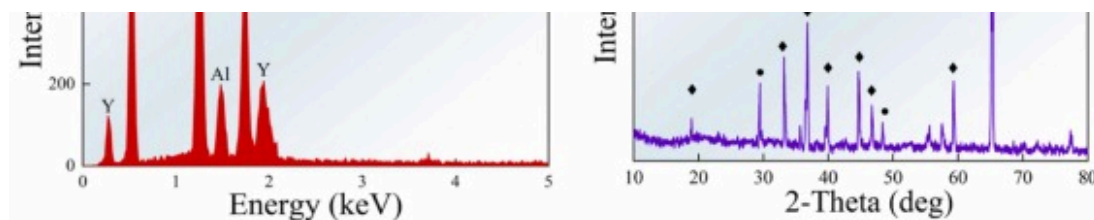
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Fig. 11. SEM images (a-b), element distribution maps (c-d), EDS patterns (e-g) and XRD pattern (h) of the BBS/YAG sample.

Fig. 12 presents the interfacial behavior between the MBS glass and YAG. There was an interface reaction between the MBS glass and YAG shown in Fig. 12(a-b). Fig. 12(c-d) show the element distribution maps of MBS/YAG sample, it can be seen that a small amount of Mg and Si elements diffused from MBS glass into the YAG substrate. Fig. 12(e-g) displays the EDS of points in Fig. 12(b) and Fig. 12(h) illustrates the XRD pattern of MBS/YAG sample. Combining the results, it was inferred that point 1 was MgAl_2O_4 phase, point 2 was Y_2O_3 phase and point 3 was glass. The Al_2O_3 from YAG substrate first reacted with the MgO in glass filler to form MgAl_2O_4 phases. The remaining Y_2O_3 was partially dissolved in the MBS glass and partially retained at the interface to form a Y_2O_3 layer, due to the low solubility of Y_2O_3 in the glass. The interfacial reaction equation is as follows:







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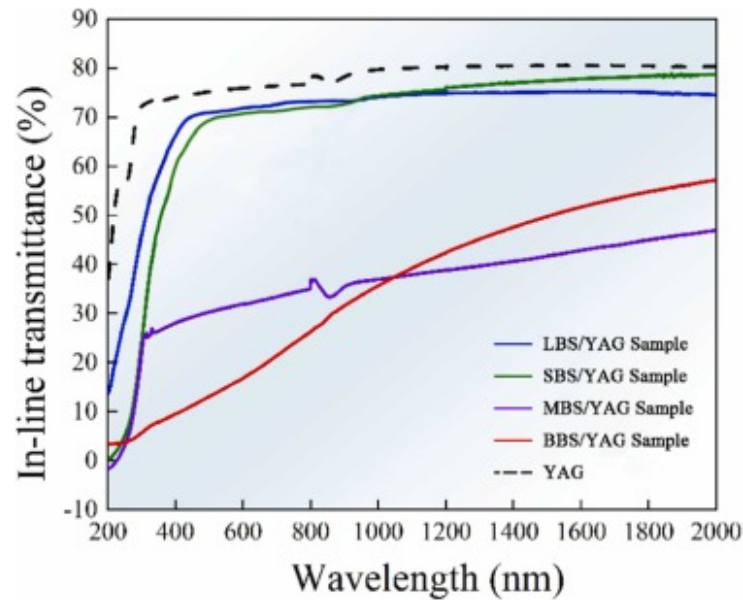
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Fig. 12. SEM images (a-b), element distribution maps (c-d), EDS patterns (e-g) and XRD pattern (h) of the MBS/YAG sample.

There are three types of interface behaviors between YAG and different glasses designed in this work involved: (1) The elements dissolved and diffused at the interface of the glass/YAG sample, but the glass remained amorphous; (2) The elements of ceramic dissolved into the glass, which changing its composition and further causing crystallization of the glass; (3) The interface reaction occurred between glass and based ceramic, leading to the formation of the interface layer.

3.4. Optical properties of borosilicate glass/YAG samples

Fig. 13 illustrates the in-line transmittance of YAG and glass/YAG samples. The in-line transmittance of the SBS/YAG and LBS/YAG samples at 1064 nm was 77.5 % and 77.1 %, respectively. They were slightly lower than the transmittance of YAG (79.9 %) at 1064 nm due to their difference in refractive index [36]. However, the in-line transmittance of the BBS/YAG and MBS/YAG samples at 1064 nm was only 41.8 % and 37.5 %. The formation of the interface layer led to the deterioration of the in-line transmission of the samples. The rapid changes in the transmittance curve of MBS glass around 800 nm may be due to the stretching vibration of $[\text{MgO}_4]$ [37].



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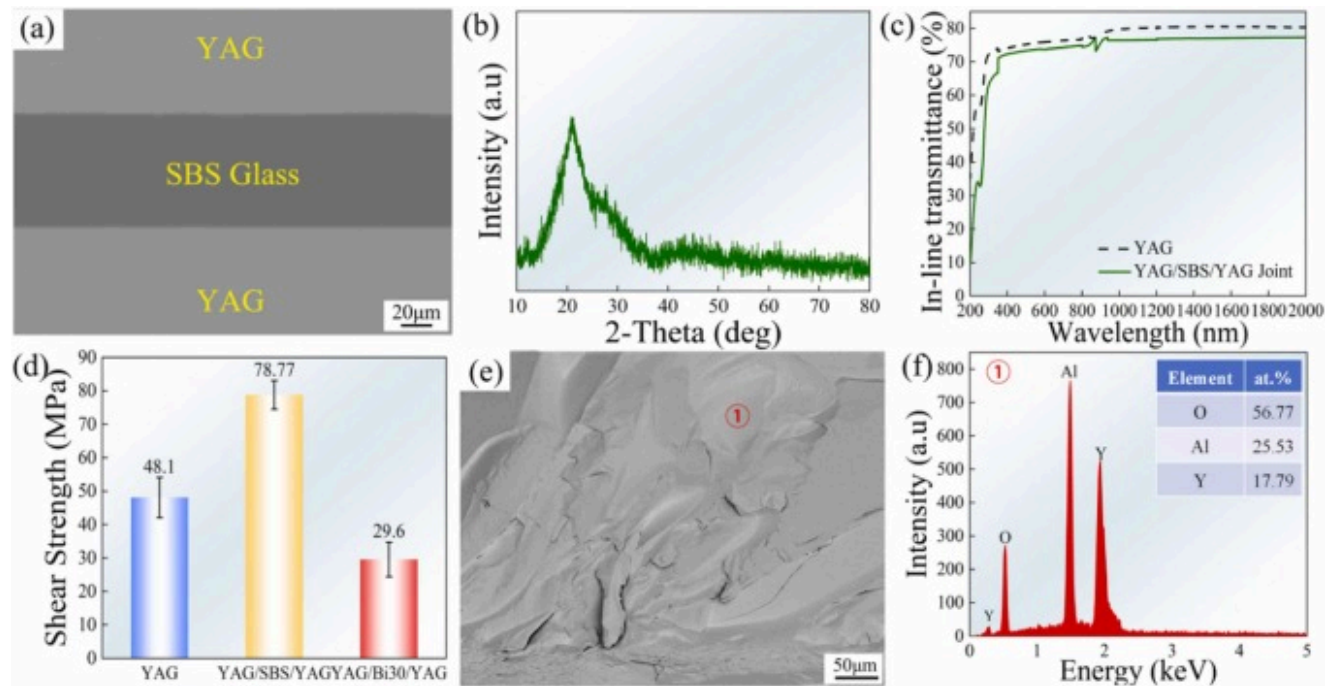
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Fig. 13. In-line transmission spectra of YAG and glass/YAG samples.

3.5. The microstructure and properties of the YAG/SBS/YAG joints

There was no interfacial reaction between the SBS glass and YAG, and the CTE of the SBS glass was the best matched with YAG. Therefore, the SBS glass was selected to bond YAG at 1450 °C for 20 min. The microstructure of the joint (Fig. 14(a)) exhibits that a straight interface was obtained and no new phase formed in brazing seam. The XRD (Fig. 14(b)) result verified that the brazing seam of joint was amorphous. Fig. 14(c) shows the in-line transmission of the YAG/SBS/YAG joint and YAG, the results indicated that the in-line transmission of the YAG (79.9 %@1064 nm) was marginally higher than that of YAG/SBS/YAG joint (76.5 %@1064 nm) probably due to refractive index of SBS glass (1.6952) mismatched with that of YAG (1.7989). The refractive index of glass refers to the extent to which light is refracted when it enters the glass from air, which is related to the composition of the glass. When light enters a medium with a different refractive index, partial reflection occurs at the interface. According to Fresnel's law [38], reflectivity is related to the difference in refractive index between two media. The greater the difference in refractive index, the greater the reflection loss, resulting in a decrease in the transmittance [39]. The average shear strength (Fig. 14(d)) of the

joints reached 78.77 ± 4.3 MPa, which was higher than the strength of the YAG/YAG joint using bismuth-borate glass filler fabricated in the literature [17]. The fracture occurred in the YAG substrate (Fig. 14(e-f)) which indicated the strength of joint equivalent to that of YAG. In summary, using the SBS glass as filler to join YAG can obtain high-strength and acceptable in-line transmittance joints.



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Fig. 14. SEM image (a), XRD pattern (b), in-line transmission spectra (c), shear strength (d), fracture morphology (e), EDS pattern (f) of YAG/SBS/YAG joint brazed at 1450 °C for 20 min.

4. Conclusion

In this work, the wetting and interface behavior between different borosilicate glasses and YAG was revealed. The YAG/YAG joint with high strength and acceptable transmittance was obtained by using the suitable glass filler. The main results are as follows:

- (1) The LBS and SBS glasses had lower crystallization activation energy than the MBS and BBS glasses. SBS glass ($7.38 \times 10^{-6}/^{\circ}\text{C}$) had the most matching coefficient of thermal expansion with YAG ($7.45 \times 10^{-6}/^{\circ}\text{C}$). All glasses had good transparency and exhibited good wettability with YAG.
- (2) Only dissolution and diffusion between glass fillers and YAG substrates existed, the glass remained amorphous at the interface. The dissolution of Al element from YAG into the BBS glass changed the BBS glass' composition and then promoted the crystallization of $\text{BaAl}_2\text{Si}_2\text{O}_8$. The MMB glass reacted with YAG to generate MgAl_2O_4 and Y_2O_3 at the interface of the MMB/YAG sample.
- (3) The SBS glass was the most suitable for bonding YAG. The YAG/SBS/YAG joint with high shear strength (78.77 ± 4.3 MPa) and acceptable in-line transmission (76.5 % at 1064 nm) was obtained.

CRediT authorship contribution statement

Ying Han: Writing – review & editing, Validation, Methodology. **Xu Ran:** Writing – review & editing, Supervision, Methodology. **Yibo Ren:** Writing – original draft, Investigation, Formal analysis. **Haohao Zou:** Visualization, Methodology, Formal analysis. **Guoqing Zu:** Methodology, Investigation, Formal analysis. **Weiwei Zhu:** Writing – review & editing, Project administration, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by National Natural Science Foundation of China (Grant Nos. [52274360](#) and [51805046](#)) and Science and Technology Development Planning Project of Jilin Province ([YDZJ202301ZYTS493](#)).

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Data availability

The data that has been used is confidential.

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